

Synaptech Education

In collaboration with: **IIT Roorkee**

Data Science + Generative AI + Agentic AI

Advanced Certification Program in Data Science, Machine Learning, GenAI & Agentic AI

**10-Months Industry-Ready
Program**



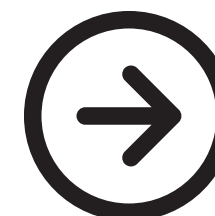


About Synaptech Education



Career-Oriented Technical Institution

Established in 2025, Synaptech Education is a premier career-oriented institution focused on Data Science & AI. In collaboration with IIT Roorkee and aligned with the Skill India Mission, we deliver practical, industry-ready learning to empower the next generation of data professionals.





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Why Data Science + GenAI + Agentic AI?



The Integrated Future-Ready Learning Model

The world's fastest-growing careers demand more than traditional data skills. By combining Data Science, Generative AI, and Agentic AI, this program prepares you for roles that are shaping the next decade — from AI engineers and ML architects to GenAI product builders and autonomous systems designers.





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Program Overview



10-Month Intensive Program Structure

Target Roles: Data Scientist, ML Engineer, AI Developer, GenAI Specialist, Data Analyst. Key Expertise: Python, ML, Deep Learning, GenAI, Agentic AI, MLOps, LLMs & RAG. Real-World Projects with industry mentorship, capstone deliverables, and placement support across 10 structured months.





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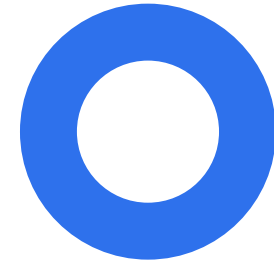
Learning Journey

Your 10-Month Transformation Path

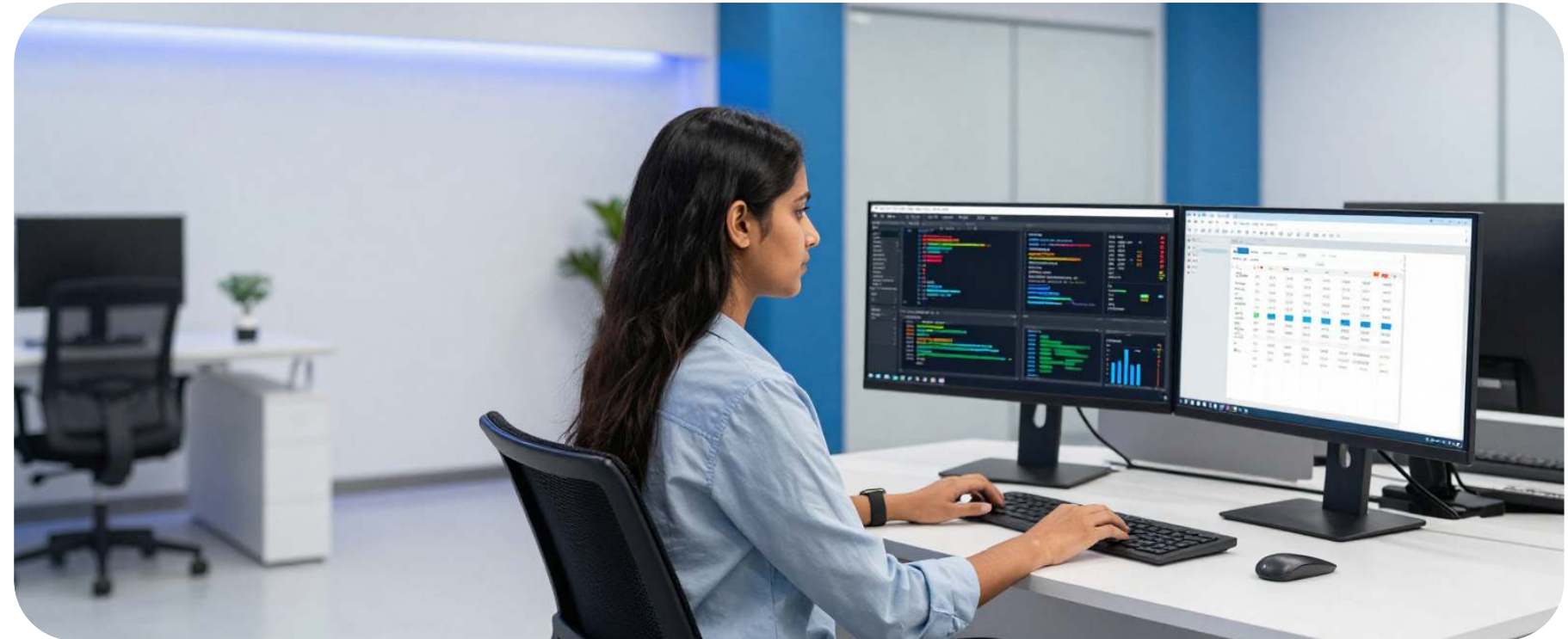
Foundation → Python & Data → Statistics & SQL → Machine Learning → Deep Learning → GenAI → RAG & LLM Apps → Agentic AI → Real-World Projects



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Data Analytics & Python



Analytics Foundations & Data Manipulation

Master Excel, SQL, Python, Power BI & Tableau through hands-on projects. Build strong data wrangling, exploratory analysis, and visualization skills to extract actionable insights from real-world datasets.





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Statistics, Mathematics & SQL



➔ Foundations of Data-Driven Thinking

Master descriptive & inferential statistics, probability theory, hypothesis testing, linear algebra, and calculus for ML. Build SQL fluency — from joins and subqueries to window functions and database design — to query, clean, and extract insights from real-world datasets.





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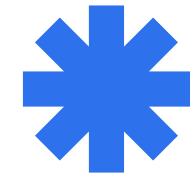
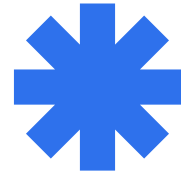


Machine Learning

→ Supervised, Unsupervised & Practical ML Workflows

Master regression, classification, clustering, and dimensionality reduction. Build end-to-end ML pipelines covering feature engineering, model selection, hyperparameter tuning, and evaluation metrics. Apply learning to real-world prediction and segmentation projects.

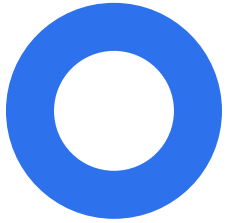
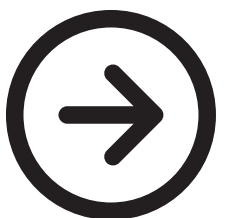




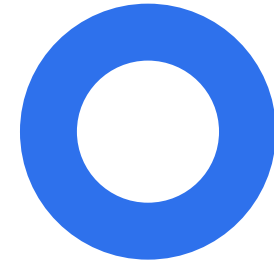
Deep Learning, NLP & MLOps

Advanced AI Engineering & Deployment

Neural networks, CNNs & RNNs for vision and sequence tasks. Transformers and attention mechanisms for NLP. Named entity recognition, sentiment analysis & text classification. Model fine-tuning, evaluation and optimization. MLOps pipelines: CI/CD for ML, model versioning & monitoring. Containerization with Docker, deployment via APIs and cloud platforms. Production-ready model serving and performance tracking.



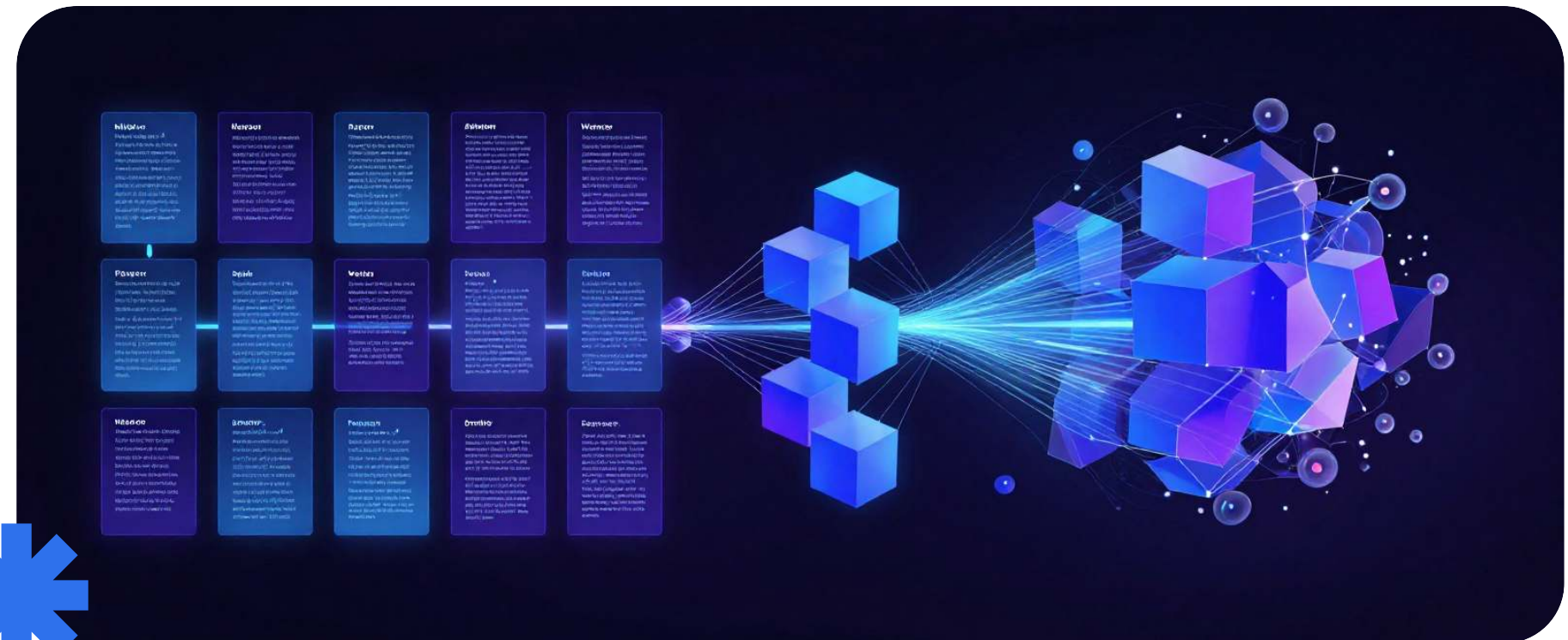
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Generative AI, LLMs & RAG

Build Intelligent GenAI Applications

Master Large Language Models, prompt engineering, and Retrieval-Augmented Generation (RAG). Explore vector databases, embeddings, and fine-tuning. Build real-world GenAI applications using OpenAI, LangChain, and cutting-edge generative AI frameworks.





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Agentic AI & Multimodal AI



Autonomous Agents. Intelligent Workflows.

Build and deploy AI agents that think, plan, and act autonomously. Master agentic workflows, multi-agent orchestration, multimodal AI systems, automation pipelines, and responsible AI practices — preparing you to lead the next generation of intelligent applications.





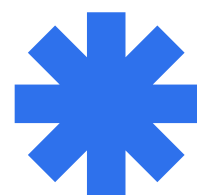
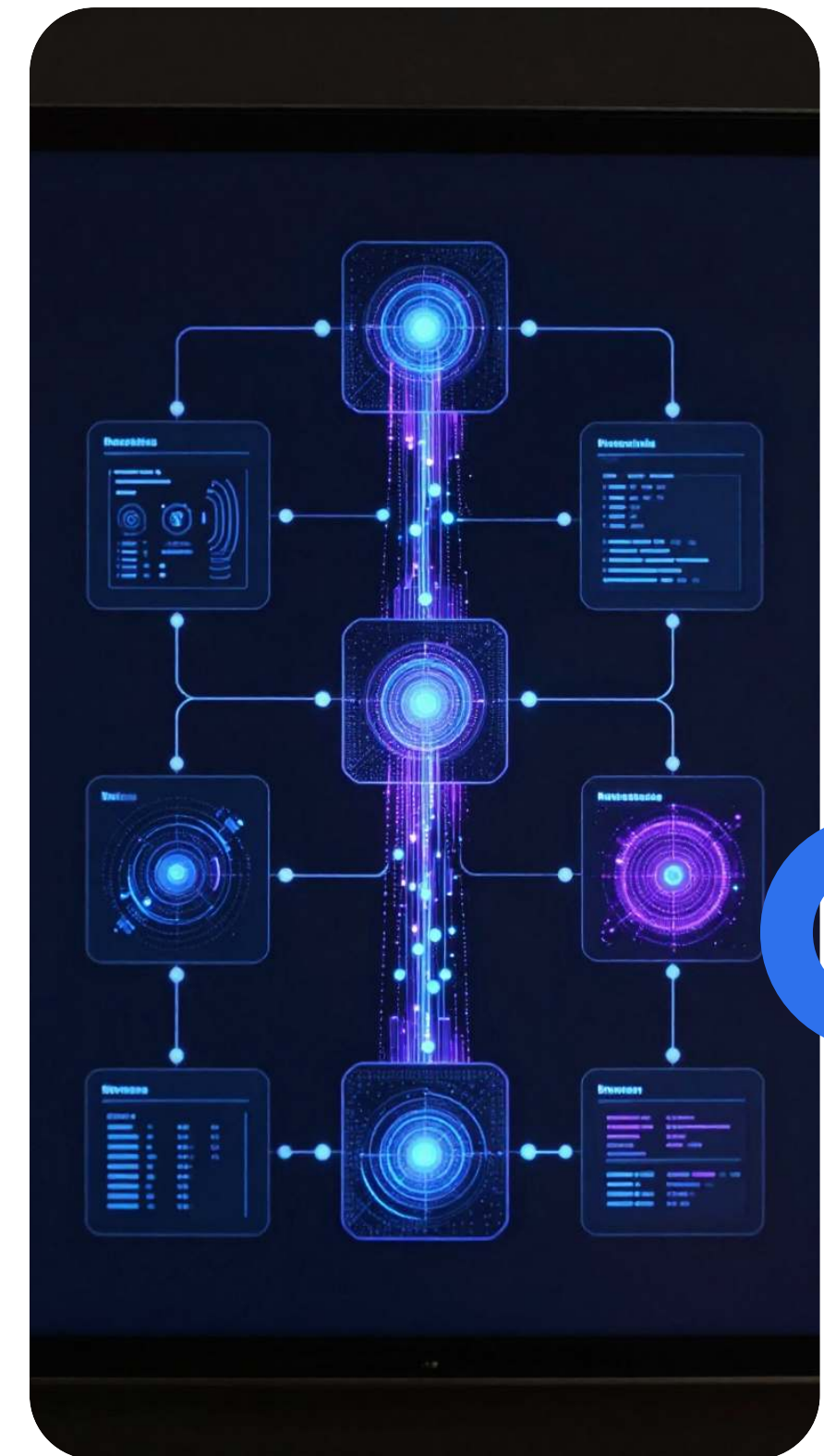
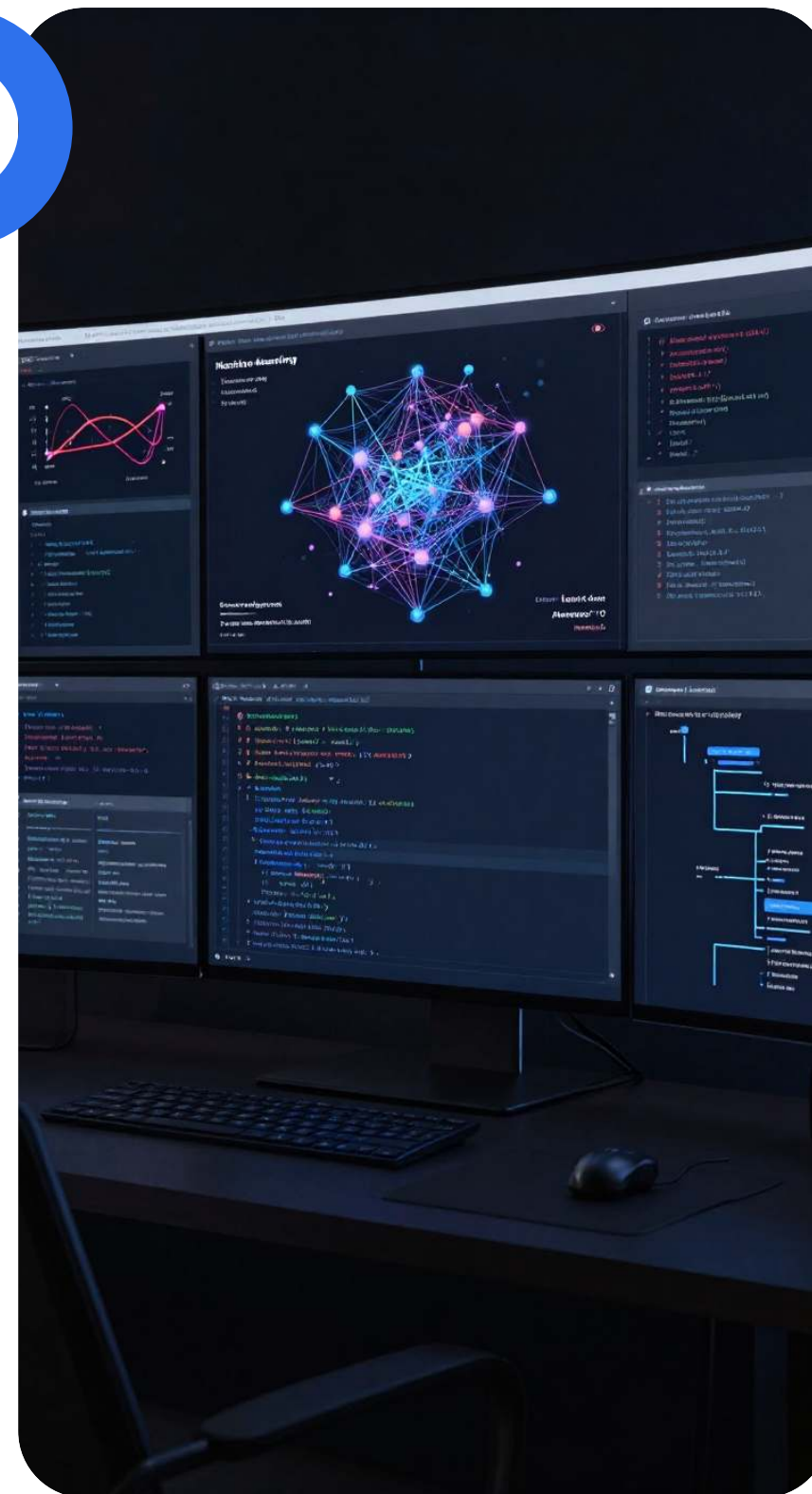
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Real-World Projects



Capstone & Industry Projects

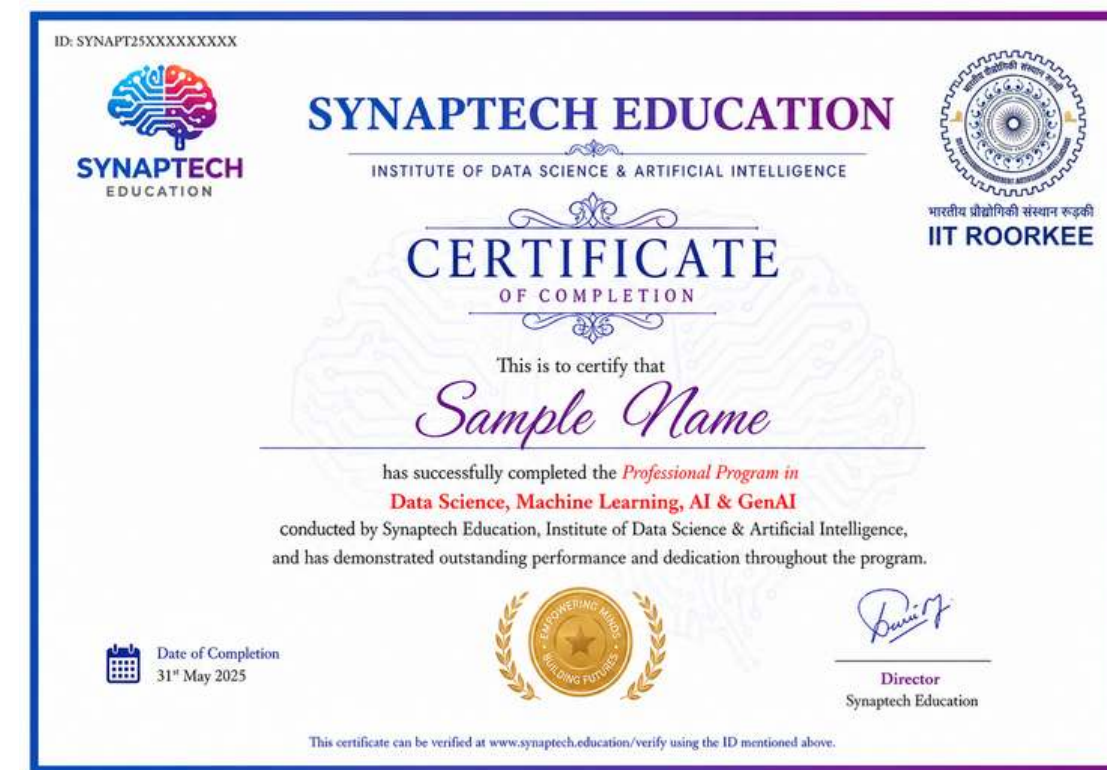
AI-Driven Drug Discovery Agent · Agentic AI Hiring & Interview System · GenAI Legal Research Copilot · Multimodal Clinical Documentation AI · AI-Powered Marketing Content Generator · RASA Chatbot · Big Mart Customer Segmentation





Synaptech Education

Launch Your Data Science Career



Certification, Internship & Placement Support

Earn dual certificates: Program Completion & Internship Certificate. Career support includes webinars, leadership sessions, 1-on-1 mentoring, mock interviews, job search assistance and LinkedIn profile building. Placement assistance with top hiring partners across AI, analytics and tech industries.



**Begin Your
AI Journey
Today**



 +91-9560940039

 www.synaptecheducation.in

 admissions@synaptecheducation.in

